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Poster: CyberSafe: Enhancing Cybersecurity Awareness

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Abstract

The rapid growth of networked embedded systems in everyday environments has significantly increased cybersecurity risks, particularly impacting young internet users. Traditional educational approaches often fail to adequately address the complex privacy and security challenges posed by these systems. Consequently, cybersecurity education leaves children underprepared for evolving privacy and security threats. In response, this research, **Cyber Safe**, introduces a gamified mobile application specifically tailored for children aged 7–10 to improve awareness of cybersecurity.

Problem Definition

There is currently a significant gap in effective online safety education for children. Many existing resources are either too complex or unengaging, and traditional teaching methods often lack the necessary interactivity. Additionally, parents and educators may not have the knowledge or resources to effectively teach these essential skills.
The goal for this research is for every child to grasp online safety fundamentals and navigate the digital world confidently. This can be achieved through engaging, age-appropriate educational tools, such as games, that teach online threats and safety practices.

Proposed Solution

The developed Android application, **CyberSafe**, integrates cybersecurity and privacy concepts specific to networked embedded systems into easy-to-use gameplay that mixes storytelling and puzzle games. To facilitate adoption by children, it is structured around four stages:

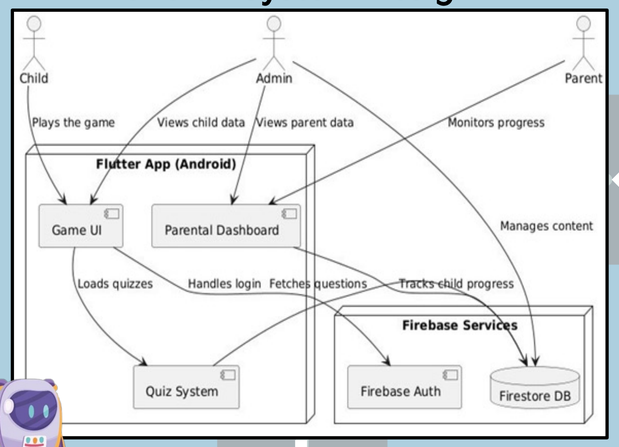
Stage 1: Protecting Personal Information.
Key topics in this stage include: digital footprint, personal data protection, privacy awareness, data collection, and secure data handling.

Stage 2: Phishing Awareness.
Key topics in this stage include: recognizing phishing attempts, identity theft prevention, secure password practices, vulnerabilities in smart toys and other embedded devices.

Stage 3: Safe Online Gaming.
Key topics in this stage include: cyberbullying prevention, safe communication boundaries, social engineering awareness, security in online gaming devices.

Stage 4: Internet of Things (IoT) Security.
Key topics in this stage include: identifying IoT devices, understanding IoT privacy settings, recognizing IoT device vulnerabilities, managing secure configurations, responding to security incidents involving IoT devices.

Overall System Design



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Preliminary Results

Child	Correct vs. Incorrect Answers	Most Understood Concepts	Most Difficult Concepts	User Engagement	Navigation and Interface Ease
Child 1	Two mistakes, then corrected them	Not sharing personal information	Identifying trustworthy people	Very engaged	Navigated easily
Child 2	3 mistakes, then corrected them	Avoiding suspicious links	Creating strong passwords	Enjoyed but slightly distracte	Needed some help
Child 3	One mistake, then corrected it	All concepts were clear	None	Very engaged	Navigated easily
Child 4	Minor difficulty in one stage, then solved it	Avoiding suspicious links and online gaming	Issue with sharing full name	Enjoyed, didn't get bored	Clearly understood navigation
Child 5	One mistake, corrected after guidance	Avoiding suspicious links and online gaming	Issue with sharing full name	Very engaged	Clearly understood navigation
Child 6	Completed all without issues	All concepts were clear	None	Very engaged	Navigated easily
Child 7	All correct	Information concepts	Navigation	Good interaction	Easy and clear
Child 8	All correct	All concepts were clear	None	Extremely enjoyable	Easy with slight difficulty
Child 9	All correct	All concepts were clear	None	Extremely enjoyable	Easy with slight difficulty
Child 10	All correct	All concepts were clear	None	Extremely enjoyable	Easy with slight difficulty

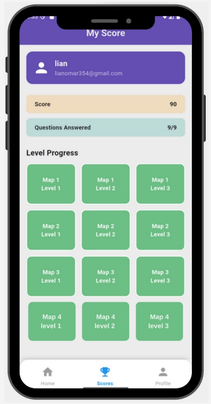
A pilot study with 10 children demonstrated a good understanding of privacy concepts, with participants showing high levels of engagement and finding the system easy to use. Minor difficulties were noted in protecting personal data, particularly when sharing full names.

Map 1: Protecting Personal Information

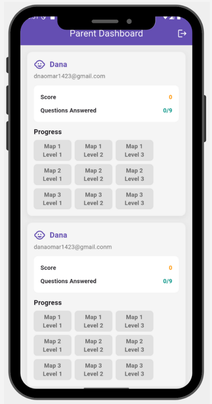
Map 2: Phishing Awareness

Map 3: Online Gaming Safety

Map 4: Internet of Things (IoT) Security



Score Page



Parent Dashboard